

India's first smart card-based PDS solution in the union territory of Chandigarh, which will involve the distribution of smart card-based ration cards to the families living below poverty line in the region. This project will capture all biometric, tax, demographic and personal information details, enabling authorities to ensure proper distribution of food rationing and other benefits.

"This project is a step in the right direction to make the PDS more robust and effective by deploying technology. The Indian government has recognised the Right to Food as a priority, and this project will ensure that the benefits of the public distribution system reach the grass root levels," says George Paul, executive vice president, HCL Infosystems.

Deployment of smart ID cards is a critical process and must be scrutinised tightly so as to meet all security issues. Paul explains that the implementation could be done in three phases, viz, develop and test the system implemented, popu-

late the system by importing data from various sources, and integrate the different systems into a single entity.

The project resonates well with Aadhaar. "The implementation of the UID project will have a catalyst effect on various e-governance projects across the country. It could play a crucial role in transforming India's image at an international level and would have major positive economical and social implications, with ample opportunity to leverage the technology far more ambitiously and intensively," says Paul.

ESIC services being spruced up

Wipro, for example, is doing a project for the Employee State Insurance Corporation (ESIC) that serves the weaker sections of the population through a network of 144 hospitals and 1388 dispensaries spread across the country.

ESIC has decided to issue smart cards to its beneficiaries, in association with Wipro. Each insured person will be given two smart cards—one for his family and the other for himself. The card will contain photos of all family members along with fingerprints for biometric authentication. By issuing such smart cards, ESIC would ensure that the insured persons can avail medical services at any ESIC facility across India, their family can also use the card even if the insured person is away, the processes are speeded up (thanks to automation), there is improved coordination across functional departments resulting in reduced operational and inventory costs, productivity of employees is enhanced by reducing redundant data entry and processes, and most importantly data is available immediately for timely decision-making.

This is expected to be one of the largest rollouts of smart IDs thus far in India.

When asked about the relevance of the UID, Lugani of Wipro says, "The UID project is a very good programme for me as it is of national interest. The project would definitely be successful if it is rolled out with the right sensitivity, and adherence and compliance by every citizen of India. This would also require every government agency across the country to honour and accept the UID Card as a sole identification of an individual."

Ought to multi-task

Over time, all government agencies would have to fall in line with this scheme. So if a person has one Aadhaar smart card with his unique 16-digit ID number on it, it should ideally be enough to authenticate the person for purposes ranging from voting to cash withdrawal. Only if the card is eventually used for multiple applications, the cost will be justified.

"India is a vast country with a huge population, and adaptation of UID is a big challenge. Lots of work needs to be done in this area. Since this is a huge effort and a lot of money is spent, the government should look at combining applications on single card. Otherwise, the same person may again have to carry multiple cards, which is again extra cost, extra effort and loss to the government exchequer," explains Shiv Turmari.

Facts and figures worth a look

- It is expected that 600 million unique ID cards, 50 million e-passports, 100 million health cards, 50 million transport and ticketing cards and 50 million banking cards will be issued over the next seven years in India.
- In India, contact-based smart cards and RFID tags currently have moderate penetration while contactless cards are yet to see a significant application base perhaps because of the cost economics.
- In select cases, the smart cards are built only with memory devices and the reader re-programs the card when connected.
- Indian telecom is the most successful application market for contact-based smart cards.
- Ticketing, toll and e-passports are the only applications tested for contactless cards.
- Several of the current government initiatives ranging from health insurance for citizens below poverty line, rural employment guarantees, voter ID issue, and ration and PAN cards can deploy smart cards.
- In India, chip suppliers and card manufacturers are technology drivers, while the market is driven by government projects. End-users seem to have limited influence in driving the need for smart cards.
- Majority of the smart cards and components (chipsets, SIM cards, readers, etc) are imported.
- Of the 140 million contact-based cards, SIM cards account for 125 million.
- Chipsets account for 40-50 per cent of the total project cost, while card, OS and integration cost account for the balance.
- Smart card chips are mainly imported from China, the European Union, Taiwan and the US.
- New technology phones like Nokia 6212C with NFC driver are capable of acting as smart cards and as a reader terminal (NFC projects implemented in Bengaluru, Andhra Pradesh and Orissa on a pilot basis).
- UID can be a backbone that drives higher deployment of smart cards.
- Strong regulations and technology convergence are likely to help achieve realistic market figures by 2014.

Courtesy: India Semiconductor Association's report titled *The Smart card Industry: A Semiconductor Industry Perspective*